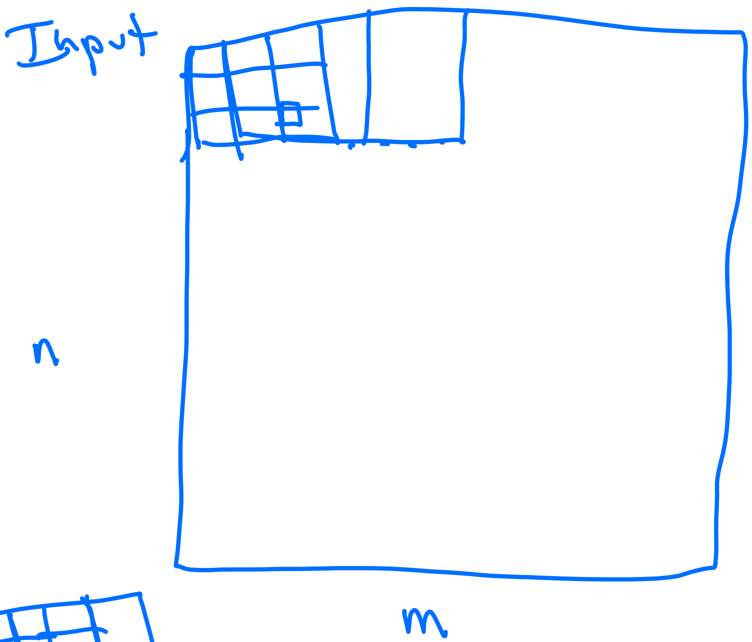
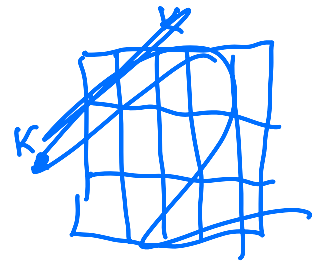
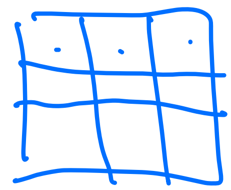
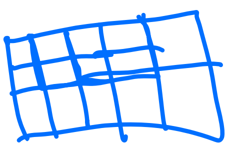


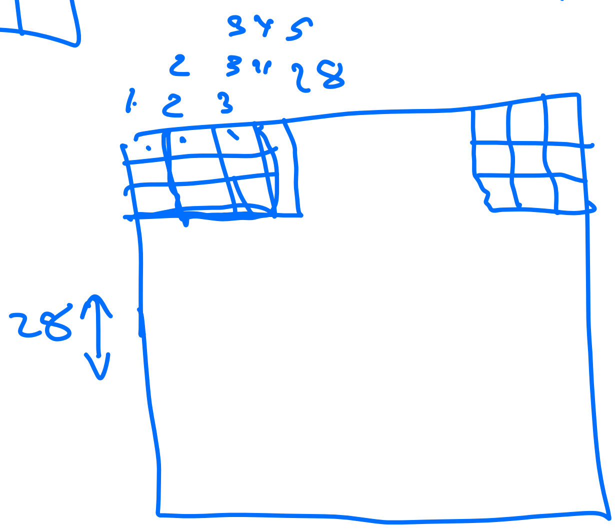


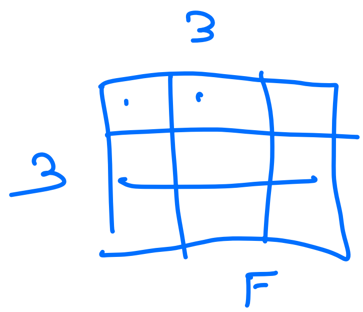


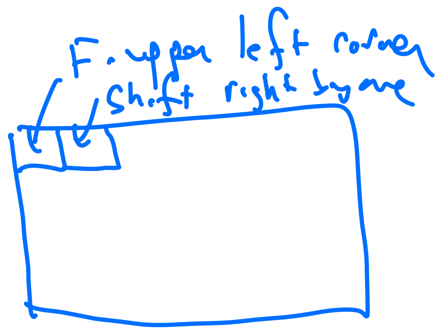
Filter



26210







3x3 filter ⟷ 28x28 image

$\Big\}$

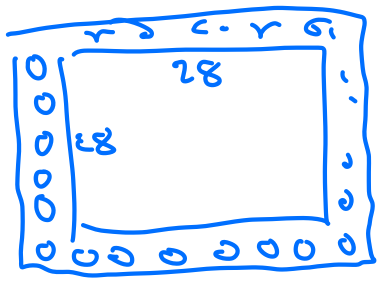
26x26 array

Image is $n \times m$

Filter is $k_1 \times k_2$

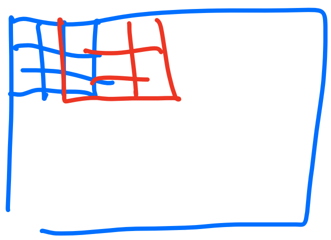
Result is $(n - k_1 + 1) \times (m - k_2 + 1)$

Padding = 1



Stride: number of rows/columns you move filter by each time.

$S = 2$



$n \times m$ image

$k_1 \times k_2$ filter

$p_1 \times p_2$ padding (by zero)

$s_1 \times s_2$

Size of result is $\left(\frac{n + 2p_1 - k_1}{s_1} + 1\right) \times \left(\frac{m + 2p_2 - k_2}{s_2} + 1\right)$

"Tensor"

L multidimensional matrix

color image: $3 \times 28 \times 28$ tensor

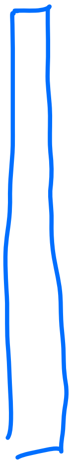
Filter: $3 \times k_1 \times k_2$ tensor

Convolution $(s = 1,\ p = 0)$

$$(3 \times 28 \times 28) * (3 \times \underset{3}{k} \times \underset{3}{k}) \rightsquigarrow 3 \times 26 \times 26 \text{ tensor } \cancel{\text{matrix}}$$

add up all 3 ⟶ single 26×26 matrix

Convolutional layer



$\boxed{\text{\# of weights: } C_{in} \cdot C_{out} \cdot f \cdot f}$

C_{in}, C_{out}, filter size $\to f$

(padding, stride)

↕

C_{out} filters of size

$C_{in} \times f \times f$

$C_{in} \times n \times m$

next layer

$C_{out} \times (n - f + 1)$

$\times (m - f + 1)$

~~6·3·5·5 weights~~

$$3 \times 32 \times 32 \xrightarrow[6 \cdot 3 \cdot 5 \cdot 5]{\text{conv}} 6 \times 28 \times 28$$

pool

$6 \times 14 \times 14$

conv ~~16·5·5~~ $6 \cdot 16 \cdot 5 \cdot 5$

$16 \times 10 \times 10$

pool

$16 \times 5 \times 5$

$\xrightarrow[\text{flatten}]{} 400 \times 1 \rightsquigarrow 10$

$400 \cdot 120$

$120 \cdot 84$

$84 \cdot 10$

$18 \cdot 25 + 95 \cdot 25 + 400 \cdot 120 + 120 \cdot 84 + 84 \cdot 10$